

Index of Flipped Classroom Video Lectures

TTK4150 – Nonlinear Control Systems
Department of Engineering Cybernetics, NTNU
(Updated on April 21, 2026)

Week 34, Lecture 1

Literature¹: Appendix A, Section 5.1

Topics: Mathematical tools, nonlinear vs linear systems

Link to youtube playlist: Lecture 1

- **L1.1: Mathematical Tools**

- 1 – Introduction
- 2 – Spaces
- 3 – Metrics
- 4 – Norms
- 5 – Inner Products
- 6 – Matrix Norms
- 7 – Differentiable Functions

- **L1.2: Nonlinear vs Linear Systems**

- 1 – Introduction
- 2 – Basic Differences
- 3 – Examples of Nonlinear Systems
- 4 – The Superposition Principle
- 5 – Invariant Sets
- 6 – Common Nonlinear Phenomena
- 7 – Common Nonlinear Phenomena (Part 2)

¹Chapters and sections refer to the textbook *H. K. Khalil, Nonlinear Systems, 3rd ed., Prentice Hall, Upper Saddle River, NJ, 2002.*

Week 35, Lecture 2

Literature: Chapter 3

Topics: Fundamental properties, second-order time-invariant systems

Link to youtube playlist: Lecture 2

- **L2.1: Fundamental Properties**
- **L2.2: Second-order Time-invariant Systems**
 - 1 – Introduction
 - 2 – Concepts
 - 3 – Phase Portraits
 - 4 – Local Analysis
 - 5 – Existence Periodic Orbits

Week 36, Lecture 3

Literature: Chapter 4

Topics: Stability of equilibrium points

Link to youtube playlist: Lecture 3

- **L3: Stability of Equilibrium Points**
 - 1 – Introduction
 - 2 – The Control Problem
 - 3 – Lyapunov Stability
 - 4 – Lyapunov Stability Analysis

Week 37, Lecture 4

Literature: Chapter 4

Topics: Lyapunov's direct method

Link to youtube playlist: Lecture 4

- **L4: Lyapunov's Direct Method**
 - 1 – Introduction
 - 2 – Motivation
 - 3 – Lyapunov Functions and Lyapunov's Stability Theorem
 - 4 – Lyapunov Theorem for Global Asymptotic Stability
 - 5 – Lyapunov Theorem for Exponential Stability

Week 45, Lecture 11

Literature: Compendium in Adaptive Control

Topics: Adaptive Control

Link to youtube playlist: Lecture 11

- **L11: Adaptive Control**

- 1 – Intro and MRAC for SISO Linear Systems

- 2 – MRAC for SISO Nonlinear Systems

- 3 – Adaptive Tracking Control for a class of MIMO Nonlinear Systems